

MEASUREMENT STANDARDS LABORATORY OF NEW ZEALAND

MEASUREMENT UNCERTAINTY GUIDELINES

A Supplement to the MSL Quality Manual

4 October 2026

Measurement Quality Council

Measurement Standards Laboratory of New Zealand

Contents

1	The GUM	1
2	Reporting expanded uncertainty	5
2.1	Rounding expanded uncertainty	5
2.1.1	Changing units	5
2.2	Unphysical limits and expanded uncertainty	6
3	Fixed-digit indications	7
3.1	Rounding	7
3.1.1	Single readings	7
3.1.2	Multiple readings	8
4	Uncertainty style	9
4.1	Background	9
4.2	Recommendations	9
4.2.1	Only the measured value is an estimate	9
4.2.2	Use the term ‘expanded uncertainty’	10
4.2.3	The level of confidence is assumed known	10
	References	11

1 The GUM

The *Guide to the Expression of Uncertainty in Measurement* (GUM) is the suite of JCGM documents providing guidance on the evaluation and expression of measurement uncertainty. The original GUM, now designated GUM Part 3, was published as a stand-alone document. Its current base edition is JCGM 100:2008, a republication of the 1995 GUM with minor corrections. Amendment 1, on nonlinearity in measurement models, was published in 2026. The former Supplements 1 and 2 are now designated GUM Parts 7 and 8, respectively.

The 2008 base edition is available free of charge from the BIPM:

<https://doi.org/10.59161/JCGM100-2008E>

Bibliographic reference The following reference is suitable for academic publications:

Joint Committee for Guides in Metrology (JCGM) 2008, *Evaluation of measurement data — Guide to the expression of uncertainty in measurement*, JCGM 100:2008, 1st edition.

GUM reference for calibration reports For the purpose of the following reference, GUM refers to GUM Part 3, JCGM 100:2008.

The following is sufficient:

See the “*Guide to the expression of uncertainty in measurement JCGM 100:2008*” (BIPM, 1st edition, 2008) for an explanation of terms.

Not all parts of the GUM suite have yet been published. Publication of a part does not itself constitute its adoption within MSL's Quality System. The table below records both the publication status of each part and MSL's current position on its use.

Table 1: Status of GUM documents within MSL's Quality System

GUM part	Old reference	Title	Publication status	MSL comment
JCGM GUM-1	JCGM 104	Guide to the expression of uncertainty in measurement – Part 1: Introduction	Published	Adopted. Replaces JCGM 104:2009. It provides an introduction and guide to the suite; it does not replace JCGM 100:2008 as the principal general method.
JCGM GUM-2		Guide to the expression of uncertainty in measurement – Part 2: Concepts	Planned	No decision required at this time.
JCGM GUM-3	JCGM 100:2008/ Amd.1:2026	Guide to the expression of uncertainty in measurement – Part 3: GUM:1995 with minor corrections	Published	Adopted. Amendment 1:2026 is to be considered separately before adoption.
JCGM GUM-4	JCGM 106	Guide to the expression of uncertainty in measurement – Part 4: The role of measurement uncertainty in conformity assessment.	Published	Not applicable to MSL activities.
JCGM GUM-5	JCGM 110 (committee draft)	Guide to the expression of uncertainty in measurement – Part 5: Examples	Published	MSL adoption decision pending.
JCGM GUM-6	JCGM 103	Guide to the expression of uncertainty in measurement – Part 6: Developing and using measurement models	Published	MSL adoption decision pending.

Continued on the following page

GUM part	Old reference	Title	Publication status	MSL comment
JCGM GUM-7	JCGM 101	Guide to the expression of uncertainty in measurement – Part 7: Propagation of distributions using a Monte Carlo method	Published	Not adopted. This is GUM Supplement 1[2]. Its use is not recommended because its interpretation of probability in uncertainty statements is Bayesian[8]. Consequently, MSL does not consider uncertainty statements obtained using its Monte Carlo method to have a physically meaningful interpretation.
JCGM GUM-8	JCGM 102	Guide to the expression of uncertainty in measurement – Part 8: Extension to any number of output quantities	Published	Partially adopted. This is GUM Supplement 2[3]. The extensions to the analytical techniques in GUM Part 3 for multidimensional measurands are adopted. The Monte Carlo material is not adopted for the reasons applying to GUM Part 7. A more extensive review of multidimensional extensions to the GUM is available [4]
JCGM GUM-9		Guide to the expression of uncertainty in measurement – Part 9: Statistical models and data analysis for interlaboratory studies	Planned	No decision required at this time.
JCGM GUM-10		Guide to the expression of uncertainty in measurement – Part 10: Applications of the least squares method	Planned	No decision required at this time.

Continued on the following page

GUM part	Old reference	Title	Publication status	MSL comment
JCGM GUM-11		Guide to the expression of uncertainty in measurement – Part 11: Bayesian methods	Planned	No decision required at this time.
JCGM GUM-12		Guide to the expression of uncertainty in measurement – Part 12: Basic methods for uncertainty propagation	Planned	No decision required at this time.

2 Reporting expanded uncertainty

2.1 Rounding expanded uncertainty

The expanded uncertainty in a measurement result should be reported to two significant figures and the associated numerical result should be reported using the same precision. Reporting fewer than two significant figures in the expanded uncertainty can significantly affect the coverage probability (level of confidence).

Reporting more than two significant figures is acceptable when there is good reason to do so. The associated result should be reported using the same numerical precision as the uncertainty.

There are situations where it may be desirable to round a value of expanded uncertainty to one significant figure. This is not recommended, but is acceptable when the least significant digit remaining is 5 or above. In such cases, rounding will introduce an error that is no more than 10 % of the unrounded value. There is no significant impact to the coverage probability in this case [5].

The practice of consistently rounding the uncertainty up is not recommended (the reason for doing this might be to 'err on the safe side' by ensuring that a conservative interval is reported). For example, 0.1234 should be not rounded to 0.13. Values should be rounded to the nearest value in the chosen number of significant figures. For example, 0.1234 rounded to two significant figures is 0.12, and 0.1251 is 0.13.

There is a NIST Good Laboratory Practice Guide [6] with examples reported to two significant figures. (The additional step of even/odd rounding in GLP 9 (2.3) is optional.)

2.1.1 Changing units

In some technical areas, a client may prefer results to be reported in units that are different from those best suited to the evaluation of measurement uncertainty. An example of this is the logarithmic units used in optical and electrical measurements.

In such cases, the usual data processing should be followed, producing a result and an expanded uncertainty in the most appropriate metrological units. Then, as a final step, the result and the limits of the expanded uncertainty interval can be transformed into the units requested by the client.

This procedure does not alter the level of confidence (coverage probability) of the expanded uncertainty. However, when a non-linear transformation is involved in the unit change, the transformed result may not be in the centre of the transformed expanded uncertainty interval, and one or other of the interval limits may seem strange (for example, a limit of infinity is

possible).

The best way of reporting this information to the client needs to be considered, as does the process by which the client can use information in the report to extract values for the standard uncertainty and degrees of freedom.

2.2 Unphysical limits and expanded uncertainty

On occasions, the expanded uncertainty calculated for a measurement result can lead to an uncertainty interval that covers unphysical values. For example, negative values for a quantity such as mass or optical density which must not be less than zero. This may happen, for example, if a measurement is sensitive to the variability in a sample of data.

Unphysical values should not generally be reported to a client. Such cases need to be carefully considered and the reason for the 'unphysical' result should be clear in terms of statistical fluctuations of influence quantities in the measurement process.

A procedure for adjusting the expanded uncertainty by removing unphysical values should be envisaged, with a clear understanding of how the 'unphysical' result arose.

In the simplest case, when a measurement model is linear, it may be acceptable to reset one of the expanded uncertainty interval limits. For instance, if an expanded uncertainty from -0.01 to 0.10 is obtained from a measurement of a quantity that must be greater than zero, the negative values may be removed to form an interval from 0.00 to 0.10. The level of confidence (coverage probability) is not changed by this operation.

A more sophisticated approach is possible for linear models when there are infinite degrees of freedom [9]. Table X in that paper describes limits of 95 % confidence intervals (expanded uncertainties), given the sample mean and sample standard deviation.

In any case, the best format for reporting to the client needs to be considered, as does the process by which the client can extract a value for the standard uncertainty and for the degrees of freedom from the information given in the report.

3 Fixed-digit indications

3.1 Rounding

The rounding guidelines in NIST Good Laboratory Practice Guide [6] apply when instrument readings are formatted in a fixed number of digits.

3.1.1 Single readings

The expression of uncertainty in a single indication should not be constrained by the format of the data.

For example, suppose an indication is limited to integer values and that the number 101 is observed. It is not recommended to report the result as 101, $U = 1$. The standard uncertainty (of a uniform distribution of error between 0 and 1) is 0.29. So with a coverage factor $k = 2$, this result would be reported as 101.00, $U = 0.58$ (or perhaps as 101.0, $U = 0.6$).

Why report more digits than the instrument can display? The extra digits are needed because the uncertainty of the reading, as an estimate of the quantity being measured, determines the number of significant figures required.

For the case of an instrument that displays truncated readings, a simple measurement model is

$$Y = w + E ,$$

where w is the reading, which is known, and E is the truncation error, which is unknown. The error can be described statistically, using a uniform distribution with zero mean and a range equal to the least significant digit displayed.

The reported value is y , an estimate of Y . This is the sum of w and the best estimate of E , which is zero. So,

$$y = w + 0 .$$

The uncertainty of y , as an estimate of Y , is found by combining the uncertainty associated with w , which is zero (we know exactly what the reading is), with the uncertainty associated with taking zero as an estimate of E , which is 0.29. So,

$$u(y) = \sqrt{(0^2 + 0.29^2)} = 0.29 .$$

3.1.2 Multiple readings

The uncertainty associated with the mean of a series of readings with a fixed number of decimal places can be evaluated by a simple method that takes advantage of variations in the data to reduce the size of the uncertainty [7]. Section 3 of Willink's paper describes the best way of processing data in this situation (a view).

This method has been independently corroborated as best-practice by NIST. It is available in GTC software as the function `type_a.estimate_digitised()`. The following example of use is in the GTC documentation:

```
# LSD = 0.0001, data varies between -0.0055 and -0.0057
>>> seq = (-0.0056,-0.0055,-0.0056,-0.0056,-0.0056,
...        -0.0057,-0.0057,-0.0056,-0.0056,-0.0057,-0.0057)
>>> type_a.estimate_digitized(seq,0.0001)
ureal(-0.005627272727272727, 1.9497827808661157e-05, 10)
```

A slightly simplified version of the result given in [7], in the case where a set of differences between a reference instrument and a device under test (DUT) is averaged to determine a correction, is to assign the uncertainty in this correction as either $s/\sqrt{N}$ or $\delta/\sqrt{12}$, whichever is larger, where s is the standard deviation of the N differences and δ is the resolution of the DUT. This is appropriate advice for second-tier calibration laboratories provided that N is at least 4.

4 Uncertainty style

4.1 Background

A measured value is an estimate of the measurand. The measurement is subject to error, and it is that unknown error that leads to uncertainty: the measured value is only an approximation of the measurand. So, for a measurand Y and some measured value of the measurand y , the standard uncertainty $u(y)$ is, strictly speaking, *the uncertainty of y as an estimate of Y* .

A combined standard uncertainty is calculated using the measurement model and the GUM rules for uncertainty propagation, with information about influence quantities.

Expanded uncertainty is of a different nature to standard uncertainty: an expanded uncertainty is used to make an inference about the measurand, whereas a standard uncertainty describes the variability of something that is inherently unpredictable, like the measurement error.

An expanded uncertainty is also qualified by a level of confidence, or coverage probability, whereas a standard uncertainty is not. There is no right or wrong value for the expanded uncertainty at a particular level of confidence.

To calculate an expanded uncertainty, one needs to know the distribution of measurement error (the default GUM assumption is Gaussian), the combined standard uncertainty associated with the result, the degrees of freedom associated with the combined standard uncertainty and the required level of confidence.

In principle, we wish to report sufficient information for clients to work out (if not provided in the report): the combined standard uncertainty, the degrees of freedom, the expanded uncertainty and the associated level of confidence.

4.2 Recommendations

4.2.1 Only the measured value is an estimate

The primary outcome of a measurement is an estimate of the measurand. So, it is unhelpful to use the term ‘estimate’ in connection with statements about values of uncertainty too. This should be avoided: it is better to say that uncertainty has been ‘calculated’, ‘evaluated’, etc, but not ‘estimated’.¹

Instead of:

These uncertainties are estimated by combining the uncertainties of the calibration process . . .

¹Many authoritative texts use ‘estimated’, but this is a bad habit that we should avoid. The MSL Guidelines on Reporting and Publishing discusses this further in the ‘Uncertainty’ section.

Prefer:

These uncertainties were calculated by combining the components of uncertainty associated with the calibration process . . .

4.2.2 Use the term ‘expanded uncertainty’

Our calibration reports refer to the current edition of the GUM for information about the terms we use. So it is unhelpful to abbreviate ‘expanded uncertainty’ unless the full term has been used elsewhere in context.

Instead of:

The uncertainty is based on a coverage factor of . . .

Prefer:

The expanded uncertainty is calculated with a coverage factor . . .

4.2.3 The level of confidence is assumed known

Given the level of confidence for a particular coverage factor, it is possible to work out the corresponding standard uncertainty and degrees of freedom (assuming a Gaussian error distribution).

The primary reason for reporting the level of confidence (or coverage probability) is to allow a client to work out degrees of freedom, so avoid language that would suggest otherwise.

Instead of:

. . . expanded uncertainties were calculated using a coverage factor $k = 2.1$ and define an interval estimated to have a 95 % level of confidence.

Prefer:

. . . expanded uncertainties were calculated using a coverage factor $k = 2.1$ for a 95 % level of confidence.

Nonetheless, if we consider the intended meaning of level of confidence, we can only expect the actual level of confidence to approximate the nominal value in practice. Situations can occur where we are confident that the actual coverage of a measurement procedure will be higher than the nominal value. For example, a value of the expanded uncertainty may be increased so that it falls within our current IANZ scope.

In such cases, do not write statements like ‘. . . at least 95 % level of confidence . . .’. Rather, simply write the report as if the expanded uncertainty were correct or include additional information to clarify how the client can use the information reported.

References

- [1] *Guide to the Expression of Uncertainty in Measurement, JCGM 100:2008 (GUM 1995 with minor revisions)*; [online](#).
- [2] *Evaluation of measurement data – Supplement 1 to the "Guide to the expression of uncertainty in measurement" – Propagation of distributions using a Monte Carlo method*, JCGM 101:2008; [online](#).
- [3] *Evaluation of measurement data – Supplement 2 to the "Guide to the expression of uncertainty in measurement" – Extension to any number of output quantities*, JCGM 102:2011; [online](#).
- [4] B D Hall, *Evaluating the measurement uncertainty of complex quantities: a selective review*, Metrologia 53 (2015) **S25**; [DOI](#).
- [5] B D Hall, *Rounding the expanded uncertainty*, Accred Qual Assur **24**, 369–373 (2019); [DOI](#).
- [6] *Good Laboratory Practice for Rounding Expanded Uncertainties and Calibration Values* (NIST, May 2019); [online](#).
- [7] R Willink, *On the uncertainty of the mean of digitized measurements*, Metrologia, 44 (2007) 73-81; [DOI](#).
- [8] D R White, *In pursuit of a fit-for-purpose uncertainty guide*, Metrologia 53 (2016) S107 – S124; [DOI](#).
- [9] G J Feldman and R D Cousins, *A Unified Approach to the Classical Statistical Analysis of Small Signals*, 1999 ; [online](#).